



Module Title

Cybersecurity and Networking

Internship Assignment Submission

Wireshark Fundamentals & Packet Analysis

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Role: Cybersecurity Intern

Duration: 3 months

Assignment Submission Date: Friday, 25th September 2026

Day:16 (Sixteen)

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1.THEORY

1.1 Answer all 20 questions in your own words

1.What is Wireshark?

Wireshark is a network protocol analyzer used to capture and inspect network packets.

2. What is packet capture?

Packet capture means recording network traffic so it can be analyzed later.

3. What is a capture interface?

Packet capture means recording network traffic so it can be analyzed later.

4. Name the three main Wireshark panes.

Packet List

Packet Details

Packet Bytes

5. What is the difference between source and destination IP?

Source IP is where the packet comes from, while destination IP is where the packet is going.

6. What does the Protocol column show?

It shows the network protocol used by the packet, such as TCP,UDP,DNS or ICMP.

7. What is a Wireshark display filter?

A display filter is used to show only specific packets from an existing capture.

8. Which filter shows ICMP traffic?

Icmp

9. Which filter shows DNS traffic?

Dns

10. Which filter shows TCP traffic?

Tcp

11. What does `tcp.port == 443` display?

It shows TCP traffic using port 443, which is commonly used for HTTPS.

12. What is DNS?

DNS translates domain names, such as `example.com`, into IP addresses.

13. What is an ICMP Echo Request?

It is a message sent by a device to check whether another device is reachable, such as when using ping.

14. What is an ICMP Echo Reply?

It is the response sent back to the device that sent an ICP Echo Request.

15. What is the TCP three-way handshake?

It is the process TCP uses to establish a connection: SYN- SYN-ACK – ACK.

16. What are SYN, SYN-ACK and ACK?

SYN: Requests to start a connection

SYN-ACK: Accepts the request and responds

ACK: Confirms the connection

17. Why is HTTPS content normally not visible as plain text?

HTTPS uses TLS encryption, so the web content is encrypted and normally cannot be read as plain text in Wireshark.

18. What is a PCAP/PCAPNG file?

It is a file format used to save captured network packets so they can be opened and analyzed later.

19. Why is Wireshark useful in incident response?

It helps security teams investigate suspicious traffic, connections, DNS activity, protocols, and network events.

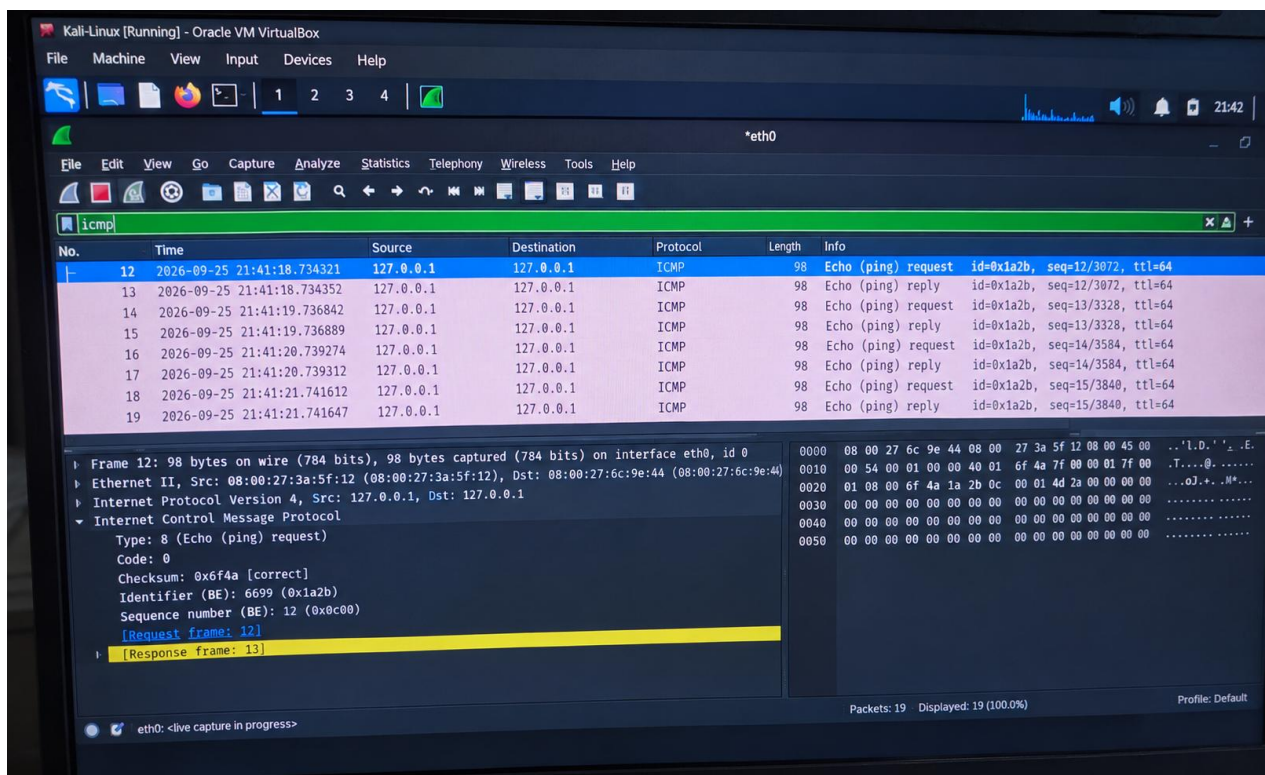
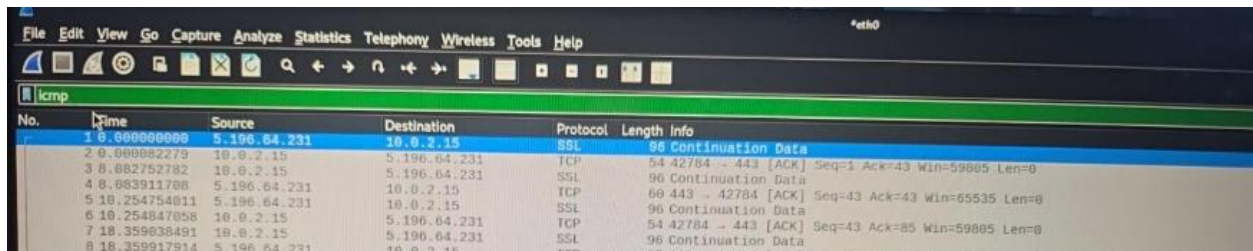
20. Why must packet capture be performed only on authorized traffic?

Because network traffic can contain private or sensitive information. Capturing someone else traffic without permission can violate privacy and security rules.

2.Lab Practical

2.1 SCREENSHOTS (your own capture only):

Wireshark interface · icmp filter · dns filter · tcp filter · tcp.port == 443 · Packet Details pane · saved day16-lab.pcapng file



Kali-Linux [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

1 2 3 4

*eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
1	18.102329	10.162.129.70	8.8.8.8	DNS	75	Standard query 0x1f27 A example.com
2	18.102412	8.8.8.8	10.162.129.70	DNS	115	Standard query response 0x1f27 A example.com CNAME ... A 93.184.216.34
3	18.102756	10.162.129.70	8.8.8.8	DNS	75	Standard query 0x8a3d AAAA example.com
4	18.102821	8.8.8.8	10.162.129.70	DNS	113	Standard query response 0x8a3d AAAA example.com AAAA 2606:2800:220:1:248:1893:25...

Frame 1: 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface eth0, id 0

Ethernet II, Src: 08:00:27:3a:5f:12 (08:00:27:3a:5f:12), Dst: 08:00:27:6c:9e:44 (08:00:27:6c:9e:44)

Internet Protocol Version 4, Src: 10.162.129.70, Dst: 8.8.8.8

User Datagram Protocol, Src Port: 53534, Dst Port: 53

Domain Name System (query)

Transaction ID: 0x1f27

Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

Queries

- example.com: type A, class IN
 - Name: example.com
 - [Name Length: 11]
 - [Label Count: 2]
 - Type: A (1)
 - Class: IN (1)

eth0: <live capture in progress>

Packets: 4 · Displayed: 4 (100.0%)

Profile: Default

Applications Places Terminal

Sep 25 21:47

*eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp

No.	Time	Source	Destination	Protocol	Length	Info
22	2.351126433	172.66.147.243	10.102.129.70	TLSv1.3	1514	Application Data
23	2.351169877	172.66.147.243	10.102.129.70	TLSv1.3	126	Application Data
24	2.351170882	172.66.147.243	10.102.129.70	TLSv1.3	126	Application Data
26	2.351178216	10.102.129.70	172.66.147.243	TLSv1.3	126	Application Data
26	2.352782226	10.102.129.70	172.66.147.243	TLSv1.3	204	Application Data
23	2.351169877	10.102.129.70	172.66.147.243	TLSv1.3	126	Application Data
24	2.351170882	10.102.129.70	172.66.147.243	TLSv1.3	126	Application Data
29	2.441003374	10.102.129.70	172.66.147.243	TLSv1.3	206	Application Data
30	2.444105381	172.66.147.243	10.106.129.70	TCP	57	443 → 36116 [ACK] Seq=6429 Ack=6365 Win=72192 Len=0 TSval=2295690624 TSecr=82806237
31	2.660378364	172.66.147.243	10.102.129.70	TCP	57	443 → 36116 [ACK] Seq=6429 Ack=6365 Win=72192 Len=0 TSval=2295690852 TSecr=82806237
32	2.663379511	172.66.147.243	10.102.129.70	TCP	57	443 → 36116 [ACK] Seq=6429 Ack=6365 Win=72192 Len=0 TSval=2295694112 TSecr=82806237
33	4.640161381	172.66.147.243	10.102.129.70	TCP	54	36116 → 443 [RST] Seq=1 Win=0 Len=0
34	4.640162640	172.66.147.243	10.102.129.70	TCP	54	36116 → 443 [RST] Seq=1 Win=0 Len=0
35	5.018592985	172.66.147.243	10.102.129.70	TCP	54	36116 → 443 [RST] Seq=1 Win=0 Len=0
36	5.018593954	172.66.147.243	10.102.129.70	TCP	54	36116 → 443 [RST] Seq=1 Win=0 Len=0
37	5.188591495	172.66.147.243	10.102.129.70	TCP	54	36116 → 443 [RST] Seq=1 Win=0 Len=0
38	5.188592840	172.66.147.243	10.102.129.70	TCP	54	36116 → 443 [RST] Seq=1 Win=0 Len=0
39	5.188645116	172.66.147.243	10.102.129.70	TCP	54	36116 → 443 [RST] Seq=1 Win=0 Len=0
40	5.188646307	172.66.147.243	10.102.129.70	TCP	54	36116 → 443 [RST] Seq=1 Win=0 Len=0
41	5.188579173	172.66.147.243	10.102.129.70	TCP	54	36116 → 443 [RST] Seq=1 Win=0 Len=0
42	5.188580352	172.66.147.243	10.102.129.70	TCP	54	36116 → 443 [RST] Seq=1 Win=0 Len=0

Frame 3: 1514 bytes on wire (12112 bits), 1514 bytes captured (12112 bits) on interface eth0, id 0

Ethernet II, Src: 08:00:27:3a:5f:12 (08:00:27:3a:5f:12), Dst: 08:00:27:6c:9e:44 (08:00:27:6c:9e:44)

Internet Protocol Version 4, Src: 172.66.147.243, Dst: 10.102.129.70

Transmission Control Protocol, Src Port: 443, Dst Port: 36116, Seq: 6429, Ack: 6365, Len: 1460

Transport Layer Security

TLSv1.3 Record Layer: Application Data Protocol: http/1.1

Content Type: Application Data (23)

Version: TLS 1.3 (0x0304)

Length: 1456

Encrypted Application Data: 5f9b7c2d1a3e4c7ff... (truncated)

eth0: <live capture in progress>

Packets: 42 · Displayed: 42 (100.0%)

Profile: Default

Kali-Linux [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

Capturing from eth0 - Wireshark

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.addr == 172.68.147.243

Wireshark - Follow TCP Stream (tcp.stream eq 4)

No.	Time	Source
15	2.451237	172.68.147.243
16	2.451280	10.102.129.70
17	2.451342	172.68.147.243
18	2.451401	10.102.129.70
19	2.451487	172.68.147.243
20	2.451553	10.102.129.70
21	2.451608	172.68.147.243
22	2.451672	10.102.129.70
23	2.451731	172.68.147.243

Frame 15: 87 bytes on wire (696 bits), 87 bytes captured
Ethernet II, Src: 08:00:27:3a:5f:12 (08:00:27:3a:5f:12)
Internet Protocol Version 4, Src: 10.102.129.70, Dst: 172.68.147.243
User Datagram Protocol, Src Port: 58432, Dst Port: 53
Domain Name System (query)
Transaction ID: 0x1a2b
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
Queries
example.com: type A, class IN
Name: example.com
[Name Length: 11]
[Label Count: 2]
Type: A (1)
Class: IN (1)

GET / HTTP/1.1
Host: example.com
User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/125.0.0.0 Safari/537.36
Accept: */*
Connection: keep-alive
HTTP/1.1 200 OK
Date: Sat, 21 Sep 2024 16:32:18 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 1256
Connection: keep-alive
Server: ECS (nyb/1D2F1)
X-Frame-Options: SAMEORIGIN
X-XSS-Protection: 1; mode=block
Accept-Ranges: bytes
<!doctype html>
<html>
<head>
<title>Example Domain</title>
</head>
<body>
<h1>Example Domain</h1>
<p>This domain is for use in illustrative examples in documents.
You may use this domain in literature without prior coordination
or asking for permission.</p>
</body>
</html>

Info
Standard query 0x1a2b A example.com
Standard query response 0x1a2b No such name A example.com
Standard query 0x3c4d A example.com
Standard query response 0x3c4d A 93.184.216.34
Standard query 0x5e6f AAAA example.com
Standard query response 0x5e6f AAAA 2606:2800:220:1:248:1893:25:
Standard query 0x7a8b A example.com
Standard query response 0x7a8b A 93.184.216.34
Standard query 0x9c0d AAAA example.com
0000 08 00 27 3a 5f 12 08 00 27 3a 5f 12 08 00 45 00E.
0010 00 53 00 1a 00 00 40 11 6c 2e 0a 02 81 46 0a 66
0020 81 ae e1 40 00 35 00 1a 2b 01 00 00 01 00 00 00
0030 00 01 00 00 00 00 03 05 78 61 6d 07 83 60 6e 65e
0040 2e 63 6f 6d 00 00 01 00 01ee.

eth0: <live capture in progress> Packets: 23 - Displayed: 23 (100.0%) Profile: Default